

Geometry-based Methods in Vision

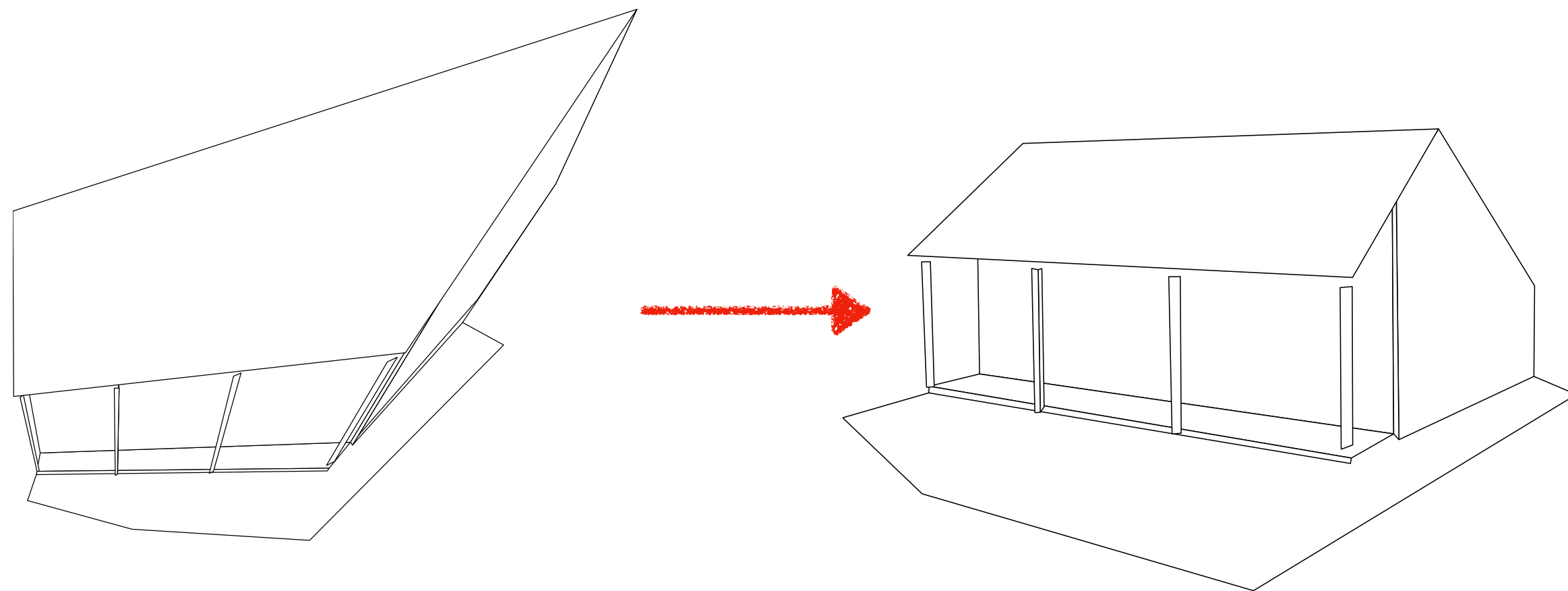
CMU 16-822, Fall 2024

Instructor: Shubham Tulsiani

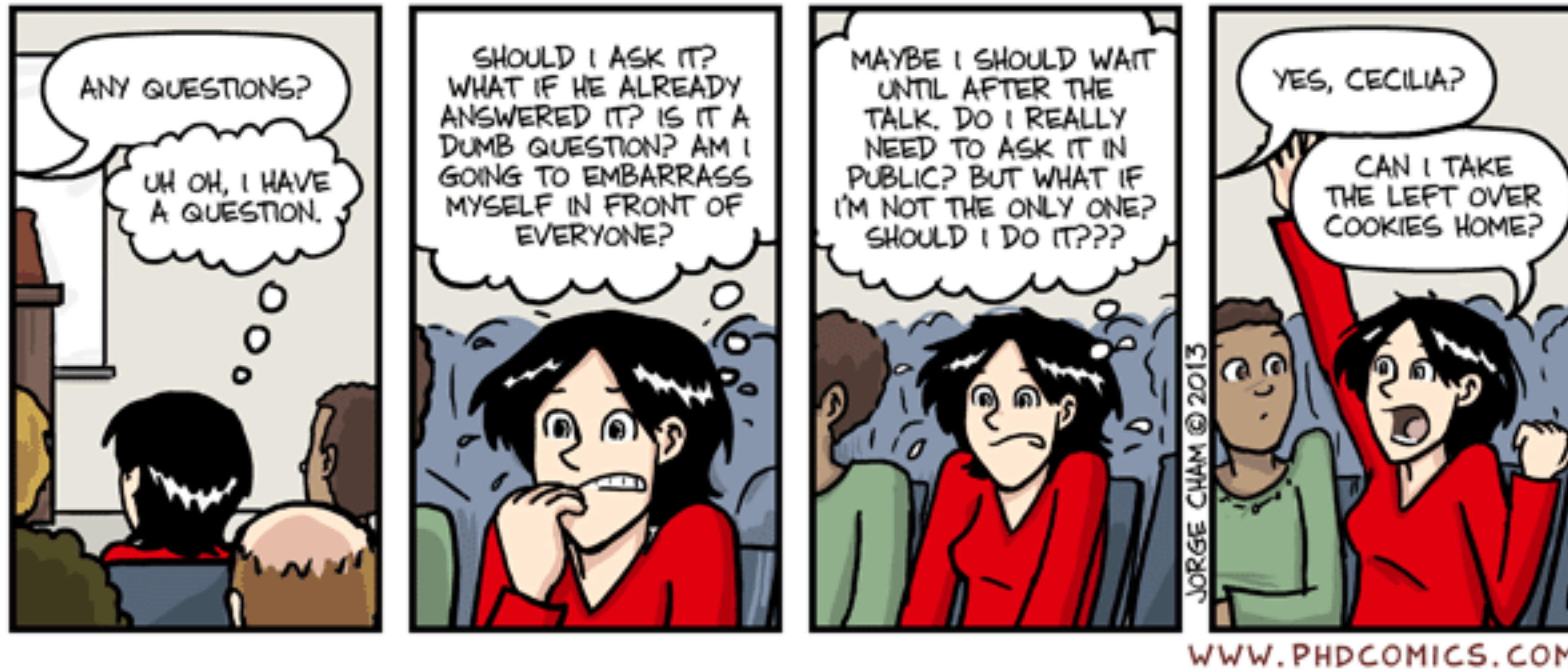
Administrivia

- Assignment 2
 - *Due on Tue next week*
- Project Proposals
 - *Due on Thu next week*

Lecture 12: Two-view Rectification



A reminder to ask questions!



Two-view Reconstruction: A simplified approach



Step 1: Annotate/extract correspondences



Step 2: Compute F (uncalibrated case) or E (calibrated case)



Step 3: Recover projection matrices P, P'



Step 4: Compute 3D via triangulation

Two-view Reconstruction



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Two-view Reconstruction: Iterative Improvement



$$\{(\mathbf{x}_i, \mathbf{x}'_i)\}$$

$$(\mathbf{P}, \mathbf{P}') \quad \{\mathbf{X}_i\}$$

So far, step-wise computation of $(\mathbf{P}, \mathbf{P}')$ and 3D points by minimizing algebraic error

Iteration: Jointly optimize cameras and 3D by minimizing **geometric** error

Important: Only optimize using the inlier matches (not all candidate matches)

Two-view Reconstruction: Iterative Improvement



$$\{(\mathbf{x}_i, \mathbf{x}'_i)\}$$



$$(\mathbf{P}, \mathbf{P}') \quad \{\mathbf{X}_i\}$$

$$\hat{\mathbf{x}}_i = \mathbf{P}\mathbf{X}_i, \quad \hat{\mathbf{x}}'_i = \mathbf{P}'\mathbf{X}_i$$

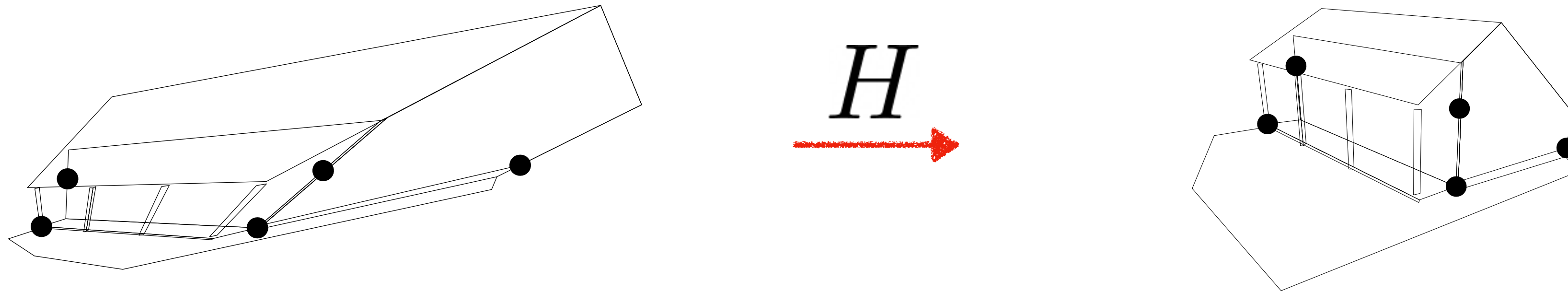
$$\min \sum_i d(\mathbf{x}_i, \hat{\mathbf{x}}_i)^2 + d(\mathbf{x}'_i, \hat{\mathbf{x}}'_i)^2$$

Q: What variables are we optimizing over?

$$\mathbf{P} = \begin{bmatrix} I & \mathbf{0} \end{bmatrix} \quad \mathbf{P}' = \begin{bmatrix} M & \mathbf{m} \end{bmatrix} \quad \{\mathbf{X}_i\}$$

assume $\mathbf{P}$ is fixed — $3N + 12$ variables

3D Rectification

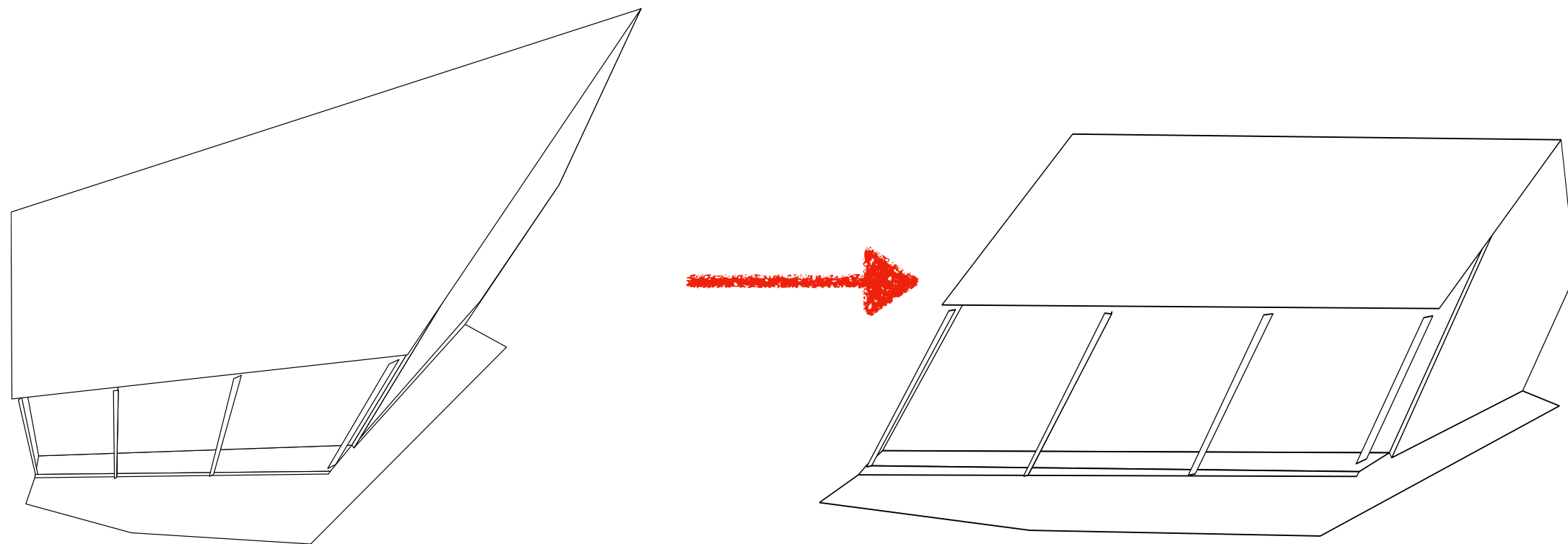
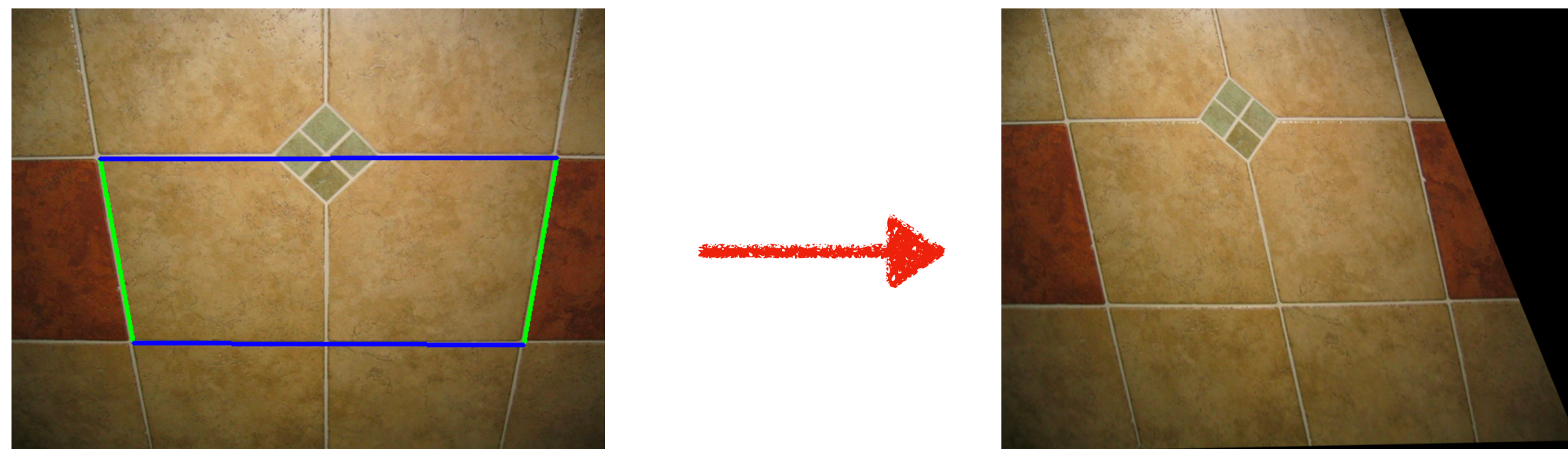


‘Direct Rectification’ — assuming known metric 3D coordinates for some points, compute H that rectifies the reconstruction

Q: How many points needed?

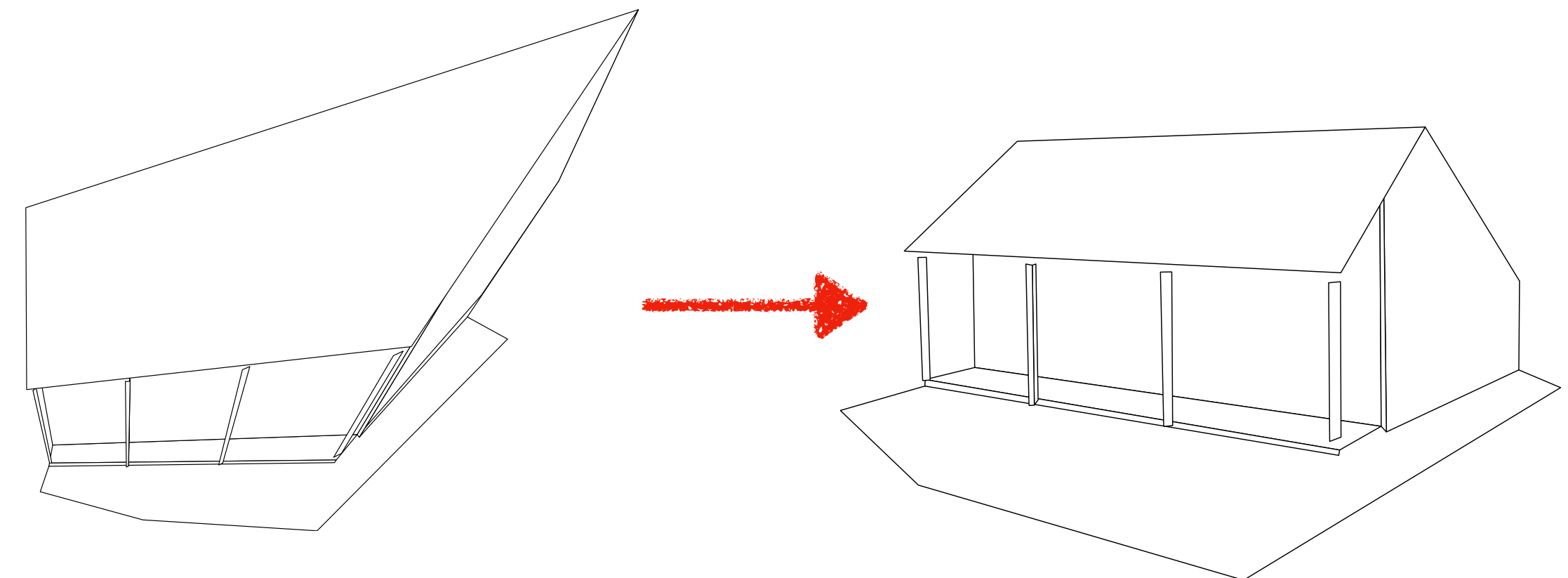
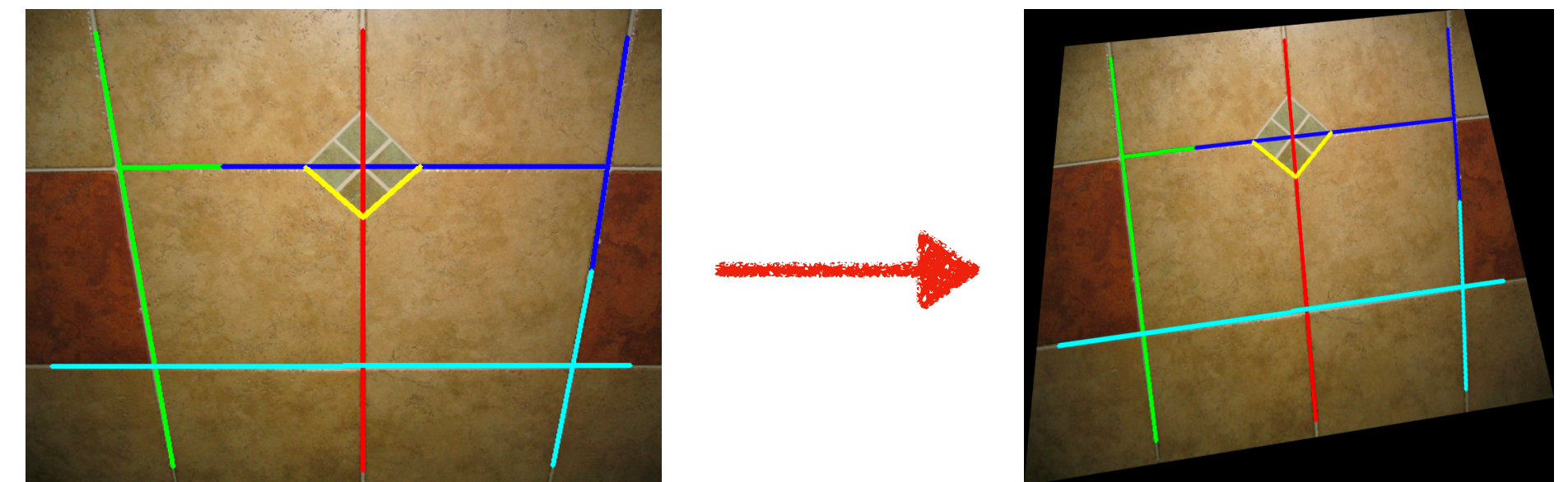
3D Rectification

Affine rectification via annotating parallel lines/directions



Compute H s.t. the line (plane) at infinity is transformed to ideal position

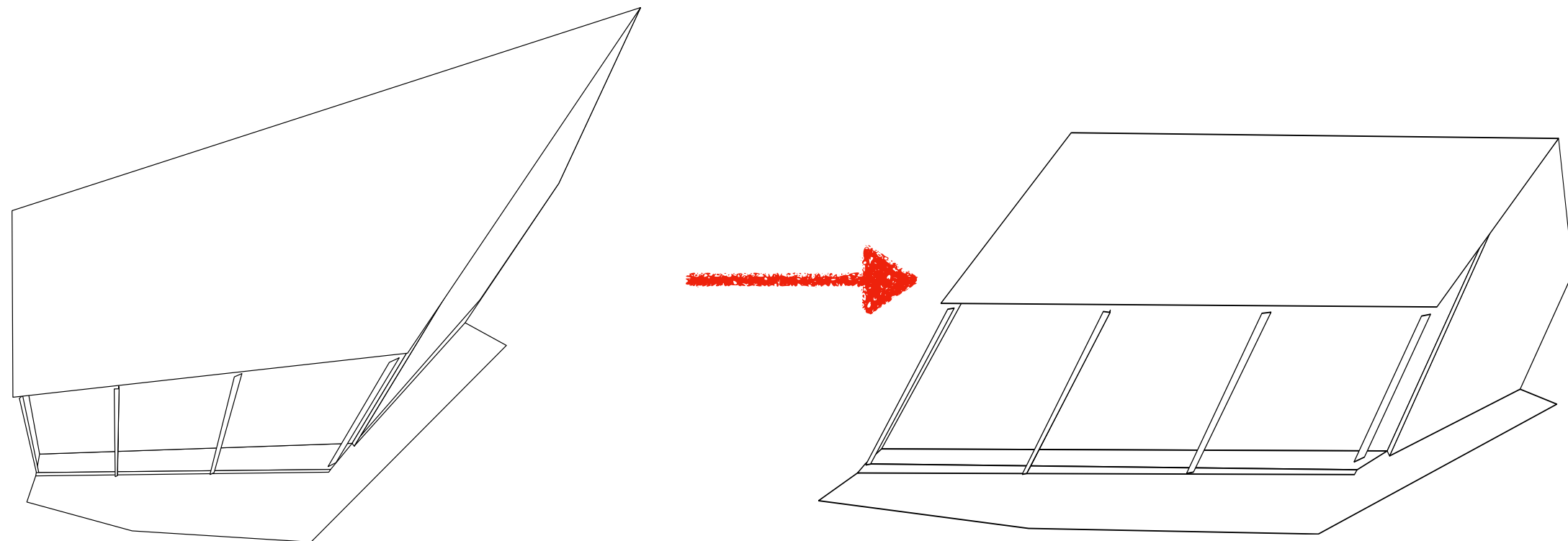
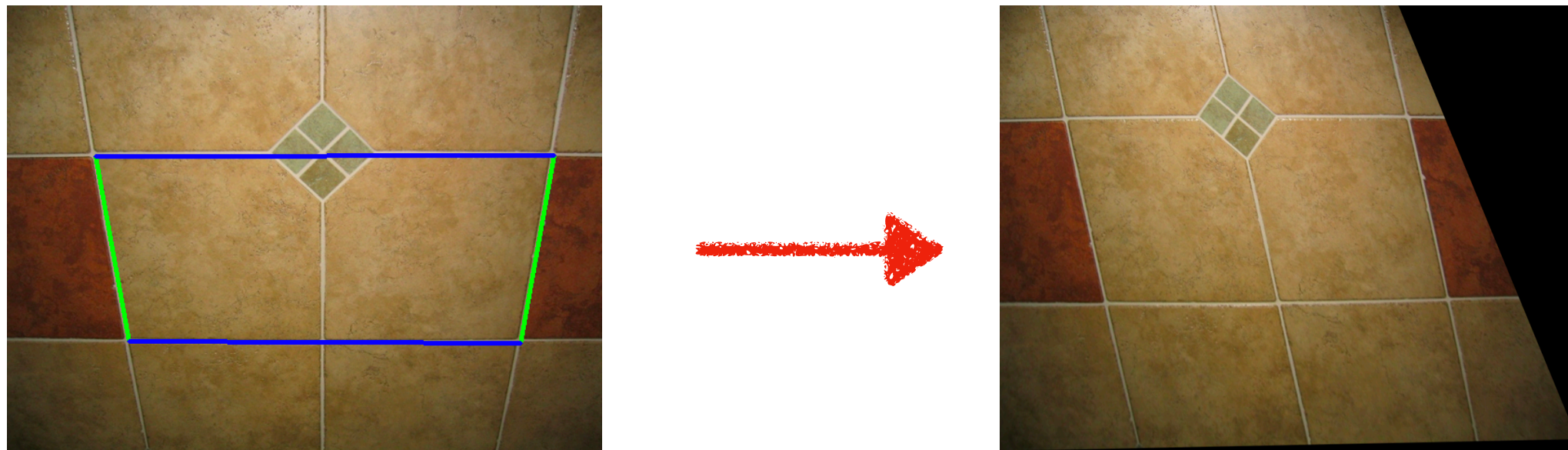
Metric rectification via annotating perpendicularity



Compute H that respects orthogonality of lines (or 3D directions)

Affine 3D Rectification

Affine rectification via annotating
parallel lines/directions



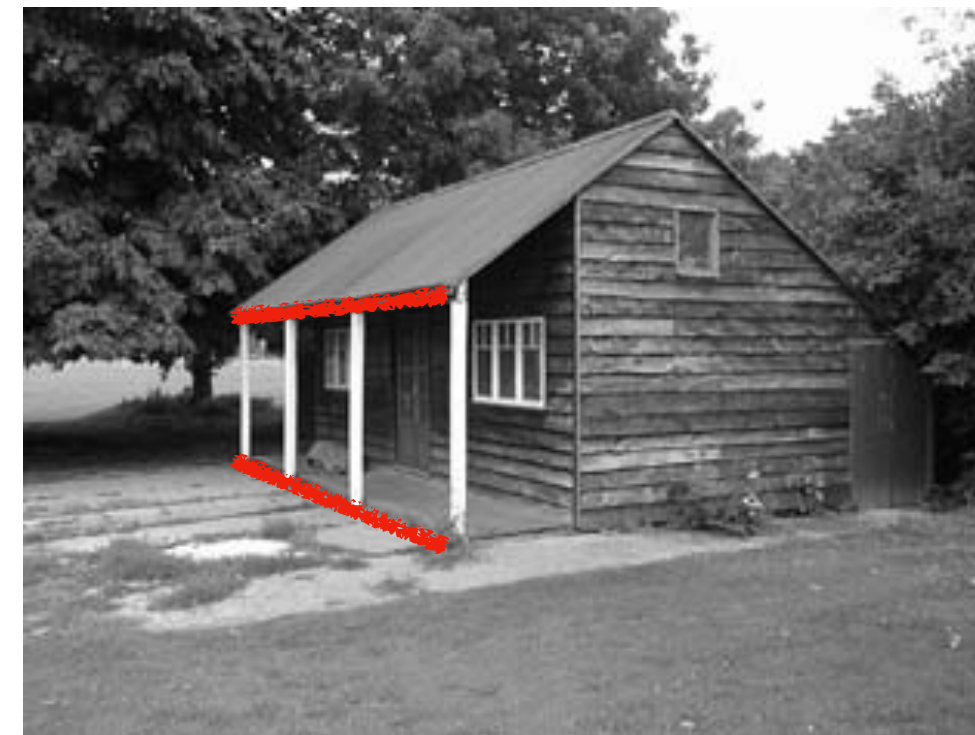
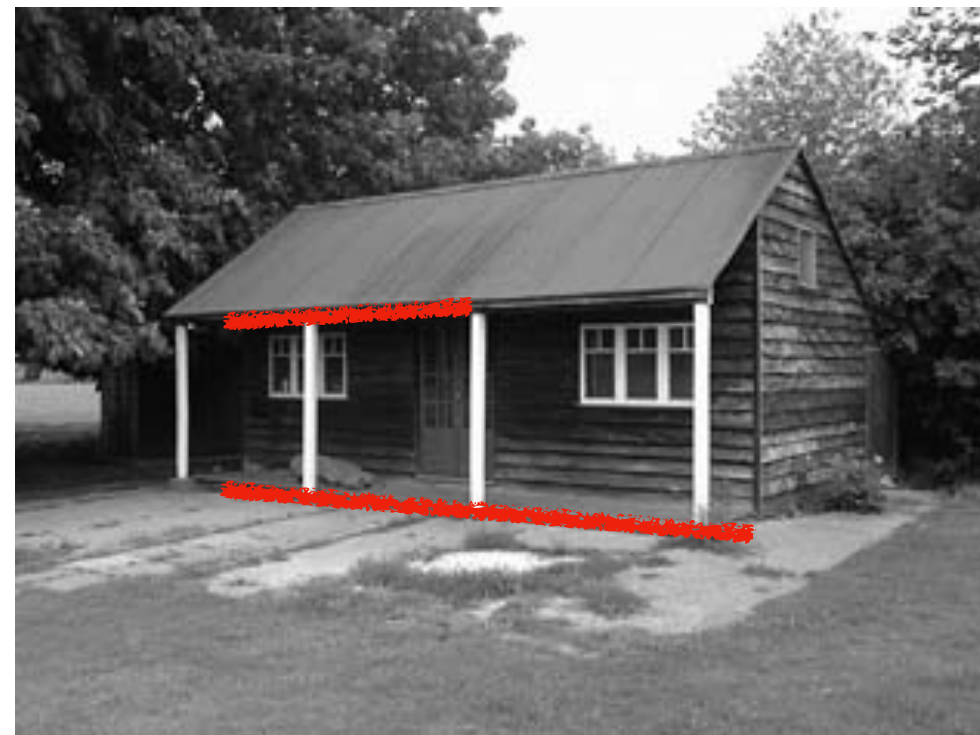
Q: How to identify the plane at infinity?

How many points needed?

Compute $\mathbf{H}$ s.t. $\mathbf{l}_{\infty} = \mathbf{H}^{-\top} \mathbf{l}'_{\infty}$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ l_1 & l_2 & l_3 \end{bmatrix}$

Compute $\mathbf{H}$ s.t. $\pi_{\infty} = \mathbf{H}^{-\top} \pi'_{\infty}$ $\begin{bmatrix} I_{3 \times 3} | \mathbf{0} \\ \pi^{\top} \end{bmatrix}$

Affine 3D Rectification: Finding Points at Infinity



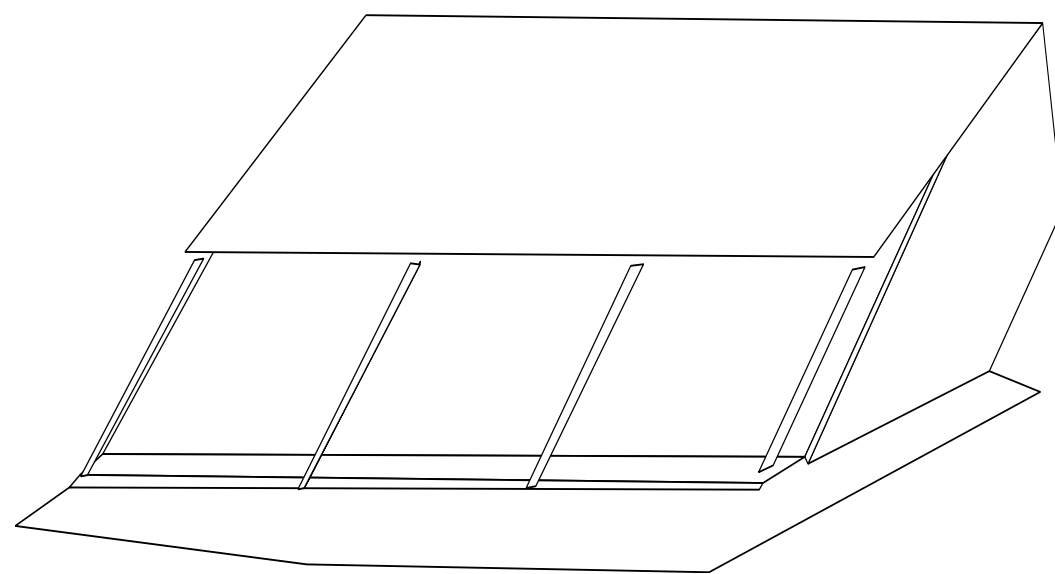
Identify corresponding vanishing points $\mathbf{v}_1$ and $\mathbf{v}_1'$

Note: can also find $\mathbf{v}_1'$ given epipolar line for $\mathbf{v}$ and just one annotated line in second image

Triangulate to find $\mathbf{x}_1$ using $(\mathbf{p}, \mathbf{p}')$

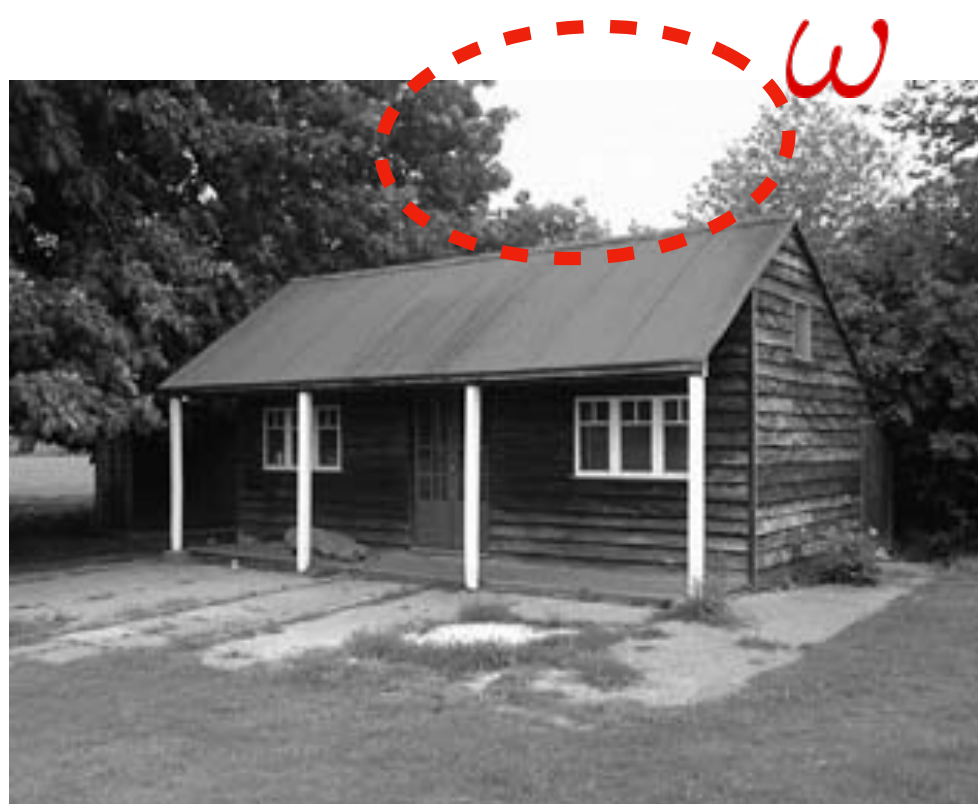
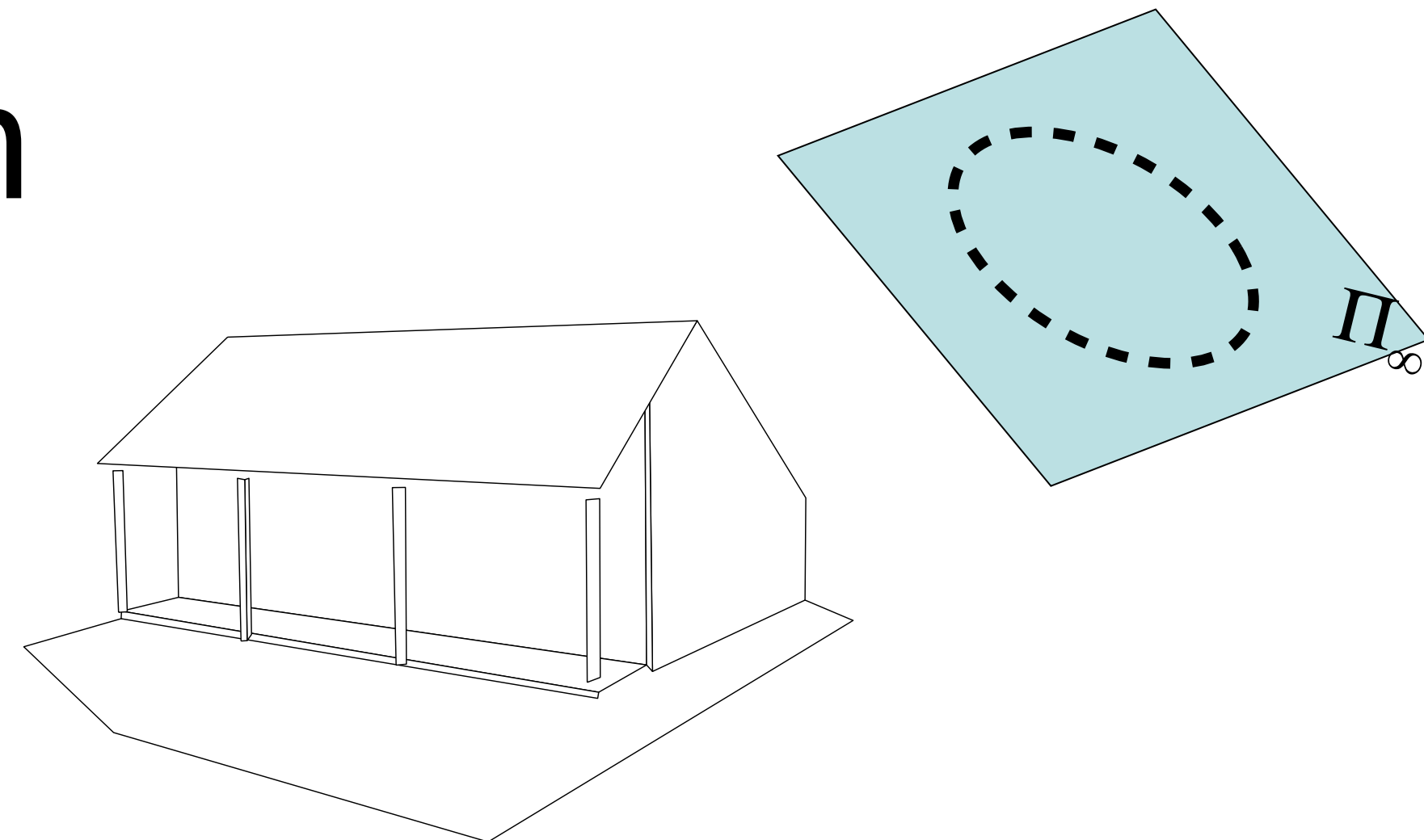
Similarly, find two other points to identify the plane at infinity

From Affine to Metric Rectification



$$\mathbf{P}H^{-1} \equiv [\mathbf{M}_H | \mathbf{m}_H]$$

$$\mathbf{M}_H \mathbf{M}_H^T = \omega^{-1}$$



If the camera is w.r.t. a metric world coordinate system:

$$\omega^{-1} = K K^T = \mathbf{M} \mathbf{M}^T$$

$$P = [\mathbf{M} | \mathbf{m}]$$

$$H^{-1} = \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix}$$

$$\mathbf{A} \mathbf{A}^T = (\mathbf{M}^T \omega \mathbf{M})^{-1}$$

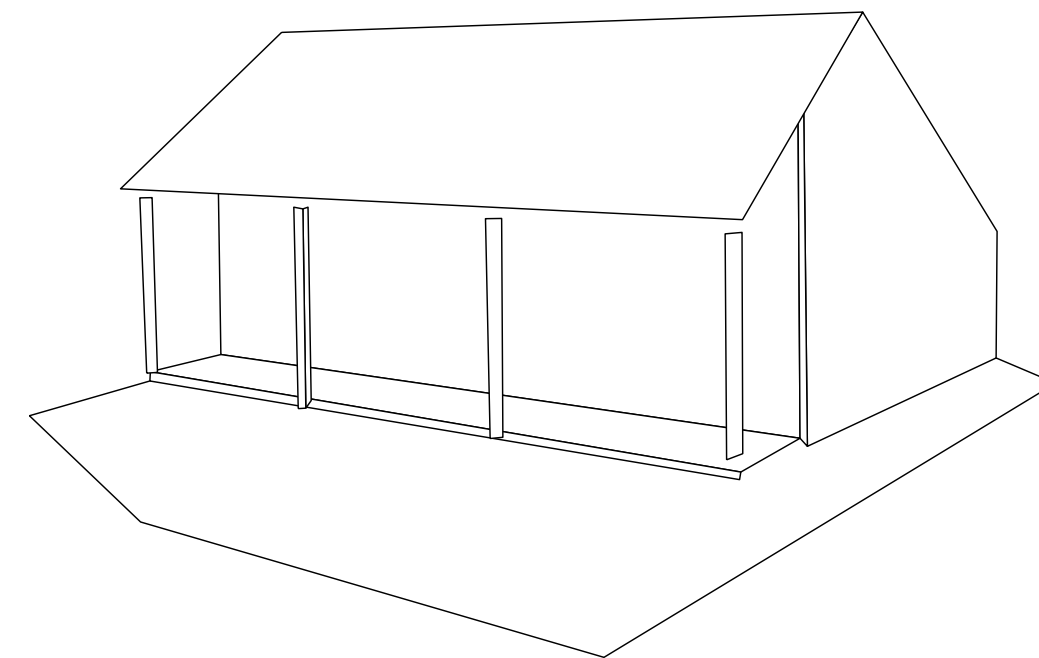
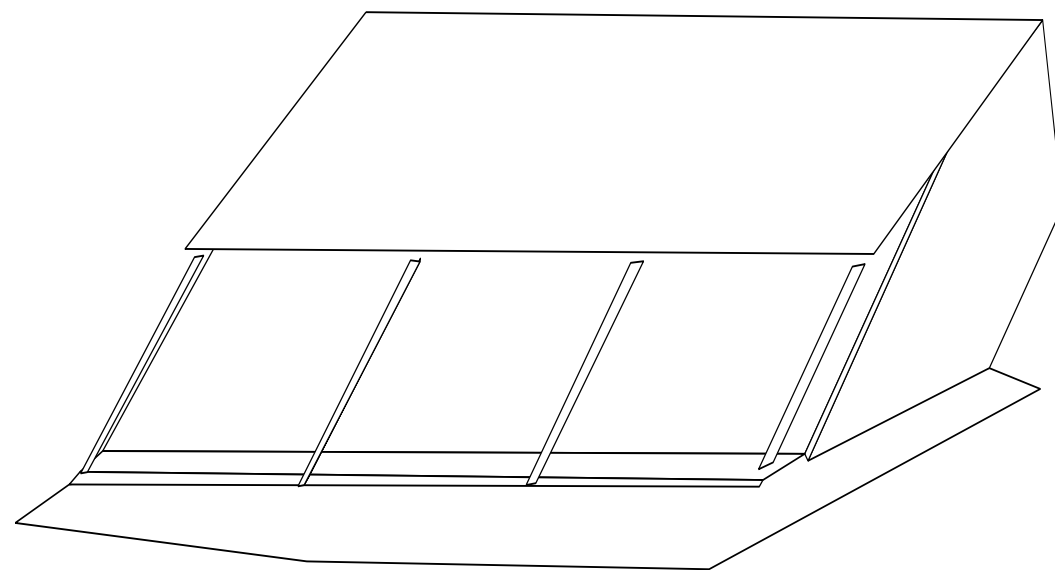
Proof: $\mathbf{M}_H = \mathbf{M} \mathbf{A}$

$$\Rightarrow \mathbf{M}_H \mathbf{M}_H^T = \mathbf{M} \mathbf{A} \mathbf{A}^T \mathbf{M}^T$$

$$= \mathbf{M} (\mathbf{M}^{-1} \omega^{-1} \mathbf{M}^{-T}) \mathbf{M}^T$$

$$= \omega^{-1}$$

From Affine to Metric Rectification



$$H^{-1} = \begin{bmatrix} A & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix}$$
$$AA^T = (M^T \omega M)^{-1}$$

Step 1: Identify IAC in some image

Step 2: Compute H as above to obtain metric reconstruction

Two-view Reconstruction



Step 1: Annotate/extract correspondences



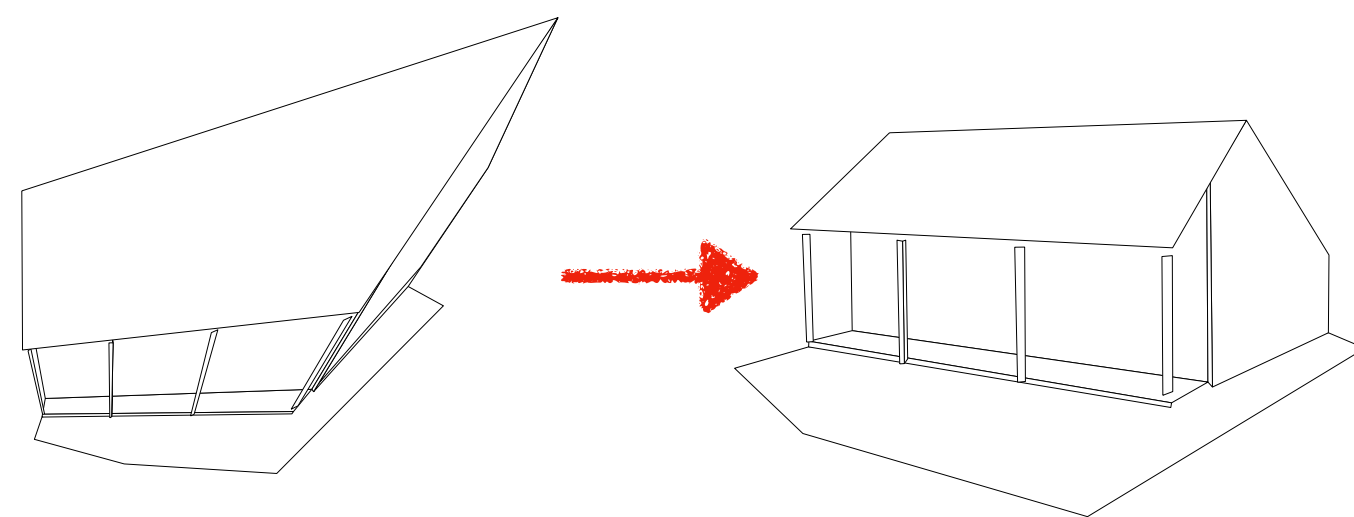
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Iterative improvement

Questions?